In the Name of God

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In the Name of God

Final Project Report: Data Analysis Course

Dataset: Flickr8k

Abstract

This project analyzes and models the multimodal Flickr8k dataset, comprising 8,091 images and 40,455 human-written captions. The principal objective is to examine the semantic structure connecting images and text and to evaluate distinct approaches to aligning these two spaces. Exploratory data analysis first examined caption-length distributions, vocabulary patterns, rare words, and visual properties. Three modeling approaches were then designed and evaluated: multimodal similarity analysis, including image-image and caption-caption neighborhoods; binary image-caption matching with a confusion matrix; and image caption generation using ResNet50 and GRU. The results showed that visual and textual spaces are not perfectly aligned, but even simple models can learn a substantial portion of the semantic structure. The caption-generation model could also reproduce frequent semantic patterns, although some examples resulted in generic captions or errors in the described action.

1. Introduction

Multimodal data are important in many machine learning and artificial intelligence problems because they describe a phenomenon in two or more different modalities, such as images and text. With such data, the challenge is not only to extract features from each modality, but also to discover the semantic relationship between their different representations. Flickr8k is a classic multimodal dataset in which every image is described by several human captions, making it particularly suitable for studying the connection between vision and language.

According to the project requirements, the work must go beyond simple implementation and include a precise problem definition, statistical data analysis, justified preprocessing, at least three distinct approaches, evaluation using multiple metrics, error analysis, bias/variance analysis, and ultimately a critical discussion of limitations. This project was designed and executed according to that framework.

1.1 Problem Definition

In this project, every image can be treated as an example in a visual space, and every caption as an example in a textual space. The main goal is to investigate whether visual and textual representations can be used to identify similarity, determine matching, and ultimately generate natural language.

Conceptually, the problem was defined at three levels:

1. Analysis of image-text alignment (similarity between captions, between images, and between images and captions)

2. Identification of matching image-caption pairs

3. Caption generation for a new image

From a machine learning perspective, the first level is primarily analytical and unsupervised, the second is a supervised binary classification problem, and the third is a supervised sequence-generation problem.

2. Introduction to the Flickr8k Dataset

The Flickr8k dataset contains 8,091 images and 40,455 captions; the captions file contains exactly five human captions for each image. The initial inspection found no missing or corrupted images, and the captions file matched the Images folder completely. This is particularly important for data validation, because any inconsistency between the image and text files could affect the training and evaluation of the models.

3. Exploratory Data Analysis

3.1 General Statistics

• Total captions: 40,455

• Unique images: 8,091

• Captions per image: 5

• Missing images: 0

• Extra images: 0

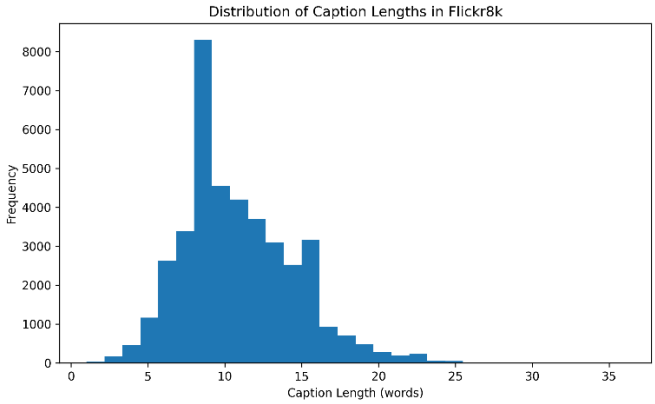
• Corrupted images: 0

These statistics indicate that the dataset is clean and systematically structured, making it appropriate for multimodal modeling.

3.2 Caption Analysis

Vocabulary analysis showed a long-tailed and highly imbalanced word distribution. Caption-length statistics are as follows:

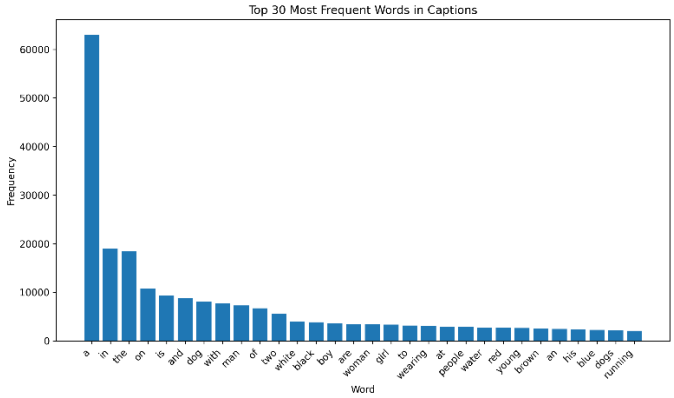
mean	10.812730194042764
std	3.767732563537824
min	1.0
25%	8.0
50%	10.0
75%	13.0
max	36.0

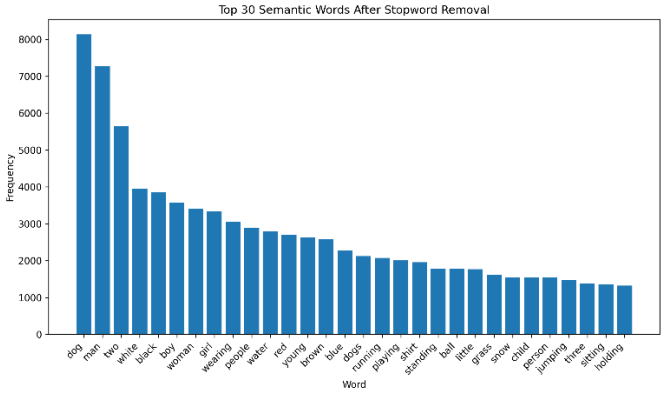


In the raw text, 437,429 tokens and 8,441 unique words were observed. After removing stopwords for semantic analysis, 250,557 tokens and 8,360 unique words remained. The most frequent semantic words included:

dog, man, two, white, black, boy, woman, girl, wearing, people, water, red, young, brown, blue, dogs, running, playing, shirt, standing.

This pattern shows that Flickr8k mainly features people, animals, outdoor activities, and descriptive colors.





3.3 Rare Vocabulary Analysis

Analysis of rare words showed that:

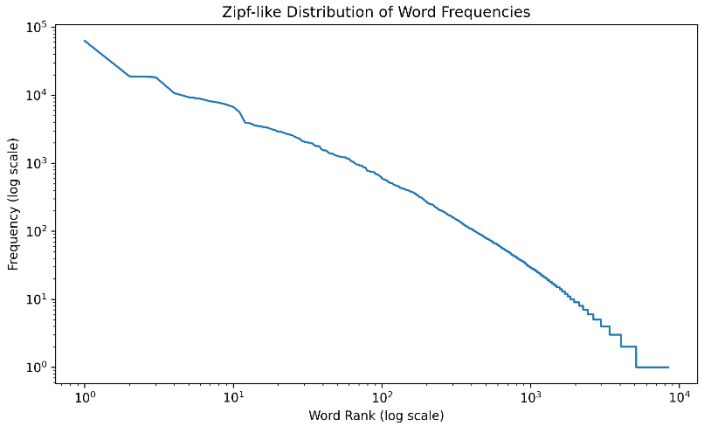
• 39.21% of the vocabulary appeared only once.

• 51.83% of the vocabulary appeared at most twice.

• 68.68% of the vocabulary appeared at most five times.

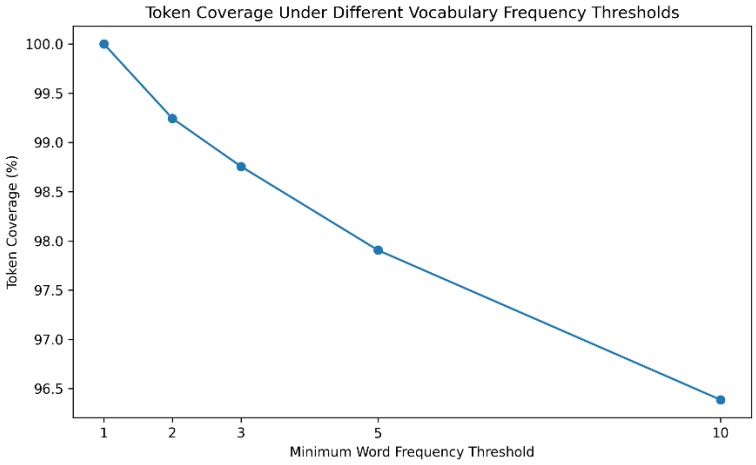
• 78.08% of the vocabulary appeared at most ten times.

	Condition	Number of words	Percent of vocabulary
0	freq = 1	3310	39.213363
1	freq <= 2	4375	51.830352
2	freq <= 3	5032	59.613790
3	freq <= 5	5797	68.676697
4	freq <= 10	6591	78.083166



This long-tailed distribution shows that a frequency threshold is essential for managing the vocabulary. Accordingly, min_freq=5 was selected for the caption-generation model because it removes a large proportion of infrequent words while still retaining approximately 97.91% of tokens.

	min_freq	kept_vocab_size	removed_vocab_size	removed_vocab_percent	kept_token_count	removed_token_count	token_coverage_percent
0	1	8441	0	0.000000	437429	0	100.000000
1	2	5131	3310	39.213363	434119	3310	99.243306
2	3	4066	4375	51.830352	431989	5440	98.756370
3	5	2971	5470	64.802748	428266	9163	97.905260
4	10	1961	6480	76.768155	421621	15808	96.386156



3.4 Image Analysis

Analysis of image dimensions showed:

• Mean width: 457.87

• Mean height: 397.25

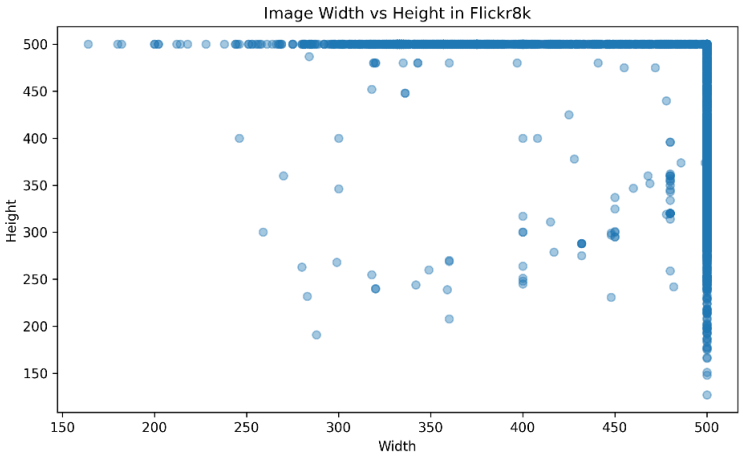
• Minimum dimensions: 127 × 164

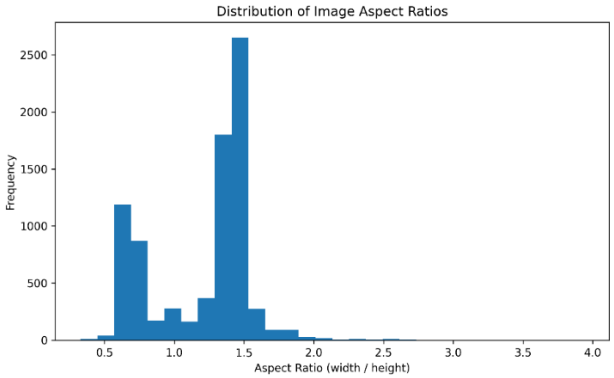
• Maximum dimensions: 500 × 500

• Mean width-to-height aspect ratio: 1.22

This variation in dimensions justifies resizing images to a fixed 224 × 224 resolution for feature extraction with ResNet50.

	width	height	aspect_ratio
count	8091.000000	8091.000000	8091.000000
mean	457.872204	397.251885	1.219682
std	68.659411	75.670168	0.361616
min	164.000000	127.000000	0.328000
25%	400.000000	333.000000	0.800000
50%	500.000000	375.000000	1.333333
75%	500.000000	500.000000	1.501502
max	500.000000	500.000000	3.937008





3.5 Reasons for Using ResNet50

ResNet50 was pretrained on the large ImageNet dataset. It has therefore learned to extract low- and high-level image features, such as edges, textures, colors, shapes, object parts, and, in deeper layers, even concepts like "dog," "person," "grass," "sky," and "outdoor scene." This is especially important for Flickr8k, because the project needs a numerical representation that reflects, to some degree, the meaning of an image rather than raw pixels alone.

Training a large CNN from scratch on 8,091 images would be time-consuming, and such a small dataset also creates a high risk of overfitting. As a pretrained model, ResNet50 allows us to use the knowledge it has already learned from millions of images. Consequently:

- Training becomes faster.
- The need for a very large dataset decreases.
- Outputs become more stable and defensible.

ResNet uses skip connections, also called residual connections. This structure helps gradients propagate more effectively and prevents the training degradation of deep networks. Since this project only requires a feature extractor, the advantage is a final representation that is generally strong and reliable.

In this project, ResNet50 was used as a feature extractor, not as an end-to-end fine-tuned model or a final image classifier. The reason is that the goal was not to classify visual categories, but to extract a semantic image representation for similarity analysis, image-text matching, and caption generation. Because Flickr8k is relatively small, fine-tuning a deep network could have increased the risk of overfitting and computational cost. Moreover, transfer learning with a new classifier is better suited to unimodal classification tasks, whereas this project is multimodal and requires a general, stable visual representation that can be connected to text. Thus, feature extraction was a balanced, computationally inexpensive, and analytically defensible choice.

Pretrained ResNet50 weights in torchvision are usually defined with a preprocessing pipeline whose standard input resolution is 224 × 224. To use those weights correctly, images should be converted to these dimensions so they match the model's original training conditions. Of course, preprocessing does not consist only of simple resizing; a standard resize-crop-normalization pipeline is usually applied. This ensures that the input follows the same statistical characteristics as those encountered by ResNet50 during its initial training.

4. Data Preprocessing

4.1 Preprocessing Decisions

• Captions were converted to lowercase.

• Punctuation was removed for exploratory analysis.

• Stopwords were removed only for semantic vocabulary analysis, not for caption generation.

• Train/validation/test splitting was performed by image to avoid data leakage.

• Previously extracted embeddings were used for the analytical and matching models.

• The captioning vocabulary was constructed from the training set only.

4.2 Data Split

The dataset was split as follows:

	Split	Unique images	Caption rows
0	Train	5663	28315
1	Validation	1214	6070
2	Test	1214	6070

There was no overlap between images in the three partitions.

4.3 Textual Features

For the analytical models, all five captions of each image were combined into a single document. TF-IDF was then applied with the following final settings:

• ngram_range = (1, 2)

• min_df = 2

• max_features = 5000

• Custom stopwords, while retaining numerical words such as two and three (common and meaningful).

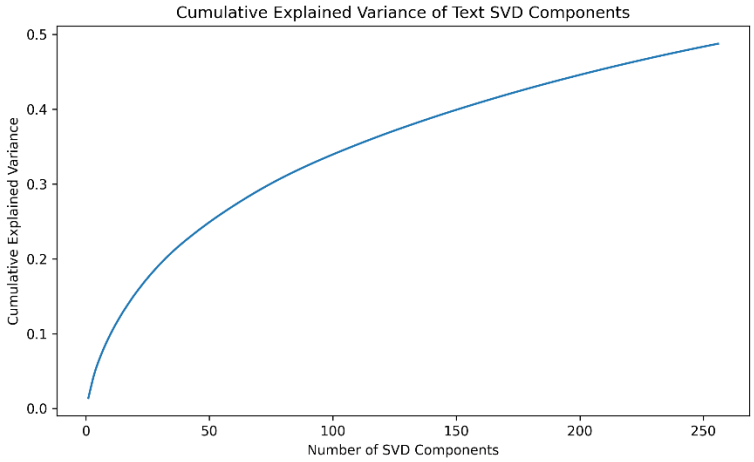
TruncatedSVD then reduced the representation to 256 dimensions, capturing about 48.76% of TF-IDF variance.

In this project, TF-IDF was selected as the textual representation. The choice of ngram_range=(1,2) allowed the model to cover both individual words (unigrams) and two-word combinations (bigrams). Single words provide basic, frequently occurring information, while bigrams preserve short, meaningful language patterns. Extending the range to trigrams would greatly increase feature dimensionality and sparsity, with little added value given the short captions in Flickr8k.

The parameter min_df=2 was chosen to eliminate very rare words appearing only once in captions. This reduces noise, preserves feature density, and focuses the representation on meaningful words. Selecting min_df=2 creates an appropriate balance between semantic coverage and noise reduction.

	ngram_range	min_df	tfidf_shape_without_cap	number_of_features_without_cap	matrix_density_percent
0	(1, 1)	1	(8091, 8360)	8360	0.225420
1	(1, 1)	2	(8091, 4734)	4734	0.388614
2	(1, 1)	5	(8091, 2578)	2578	0.686139
3	(1, 2)	1	(8091, 101896)	101896	0.044129
4	(1, 2)	2	(8091, 28433)	28433	0.126211
5	(1, 2)	5	(8091, 9124)	9124	0.327500
6	(1, 3)	1	(8091, 283044)	283044	0.025710
7	(1, 3)	2	(8091, 45854)	45854	0.094769
8	(1, 3)	5	(8091, 11651)	11651	0.283727

The limit max_features=5000 was also imposed to control the final dimensionality of the TF-IDF matrix. It reduces memory use and computational expense while allowing dimensionality reduction with SVD. These settings are consistent with the actual data and parameter-sensitivity analysis, and they allow later models, including image-caption matching and caption generation, to use a dense, meaningful numerical representation of text.



4.4 Visual Features

For images, pretrained ResNet50 was used as the feature extractor, producing a 2,048-dimensional vector per image. For some models, PCA then reduced this to 256 dimensions. PCA preserved approximately 77.01% of the variance in the visual vectors.

5. First Model: Multimodal Similarity Analysis

5.1 Objective of the First Model

This approach aimed to investigate semantic structure both within and between modalities, revealing relationships between images and captions and the internal semantic structure of images and text. In particular, three measures were studied:

Similarity between captions (caption-caption similarity)

Similarity between images (image-image similarity)

Overlap of nearest visual and textual neighbors (visual-text top-5 neighbor overlap)

5.2 Method

The following standard approaches were used to analyze similarity:

Caption-caption similarity: Text vectors were extracted with TF-IDF and reduced to 256 dimensions using TruncatedSVD. Similarity between vectors was measured using cosine distance.

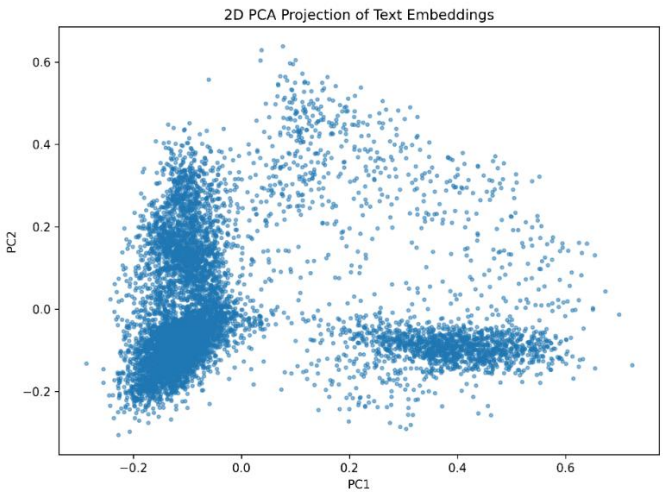


Image-image similarity: 2,048-dimensional visual features were extracted using pretrained ResNet50 and then reduced to 256 dimensions with PCA. Cosine distance was used to measure image similarity.

Visual-text consistency: For each image, its top five nearest visual neighbors and top five nearest textual neighbors were identified; the overlap between the two sets was then calculated.

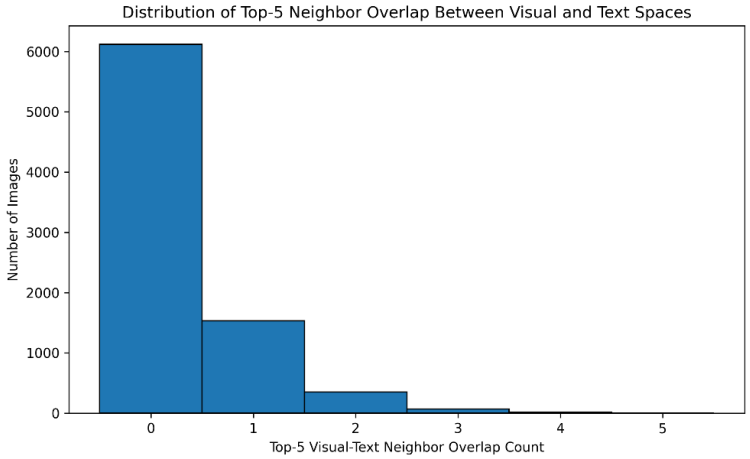
Together, these methods make it possible to analyze semantic relationships between image and text spaces and the internal structure of each space.

5.3 Results

Mean overlap between top-5 visual and textual neighbors: 0.3103

Mean overlap ratio: 0.0621

Median overlap: 0



	visual_text_neighbor_overlap	overlap_ratio
count	8091.000000	8091.000000
mean	0.310345	0.062069
std	0.614988	0.122998
min	0.000000	0.000000
25%	0.000000	0.000000
50%	0.000000	0.000000
75%	0.000000	0.000000
max	5.000000	1.000000

These results show that for many images, the nearest visual and textual neighbors are not identical, although a meaningful proportion of pairs are semantically close.

One example with five overlapping visual and textual neighbors is shown below:



A group of people sit around a blazing fire at night.

5.4 Interpretation

Visual and textual spaces are not fully aligned: ResNet50 mainly captures visual features such as shape, texture, and composition, whereas TF-IDF emphasizes vocabulary overlap.

Nevertheless, successful cases showed that semantically similar images and captions appear close together. For example, images of dogs, grass, running, and jumping were clustered well in both the visual and textual spaces.

This analysis indicates a relative semantic correlation between the visual and textual spaces, although that correlation is neither linear nor perfect.

The model showed that Flickr8k has a coherent structure in both text and vision, but the relationship between the two modalities is not perfectly linear. Multimodal visual-textual similarity analysis provides a foundation for subsequent models, an overall view of semantic clusters and spatial overlap, and insight into the limitations of separately constructed embeddings.



6. Second Model: Image-Caption Matching

6.1 Objective of the Second Model

The second approach reformulated the task as binary classification: do an image and a caption belong together or not?

6.2 Constructing the Training Data

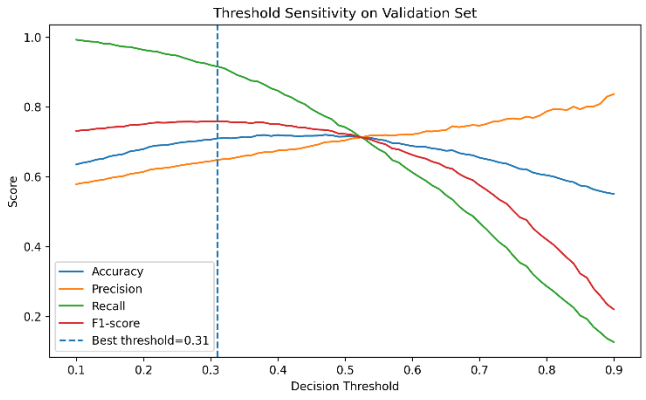
For each image, a positive pair was created from the image and one of its genuine captions. To create a negative pair, that same image was paired with a caption belonging to a different image. This produced aligned, balanced training data. In the train, validation, and test partitions, one positive and one negative pair were created per image, preserving a 1:1 class ratio.

6.3 Pair Feature Representation

Visual features were extracted using ResNet50 and reduced to 256 dimensions by PCA. Text features obtained from TF-IDF+SVD were likewise transformed into 256 dimensions. For each image-caption pair, a combined feature vector [v, t, |v - t|, v * t] was then constructed to examine both similarities and differences.

6.4 Classification Model and Threshold Selection

Logistic Regression was used for classification. The model was first trained on the training set, after which the probability of a match was calculated for the validation set and the threshold was chosen to maximize the F1-score. The optimal threshold was 0.31. F1 is appropriate here because both precision and recall matter; focusing on accuracy alone could be misleading.



6.5 Results

At threshold=0.31, the model achieved the following test results:

• Accuracy ≈ 0.6998

• Balanced Accuracy ≈ 0.6998

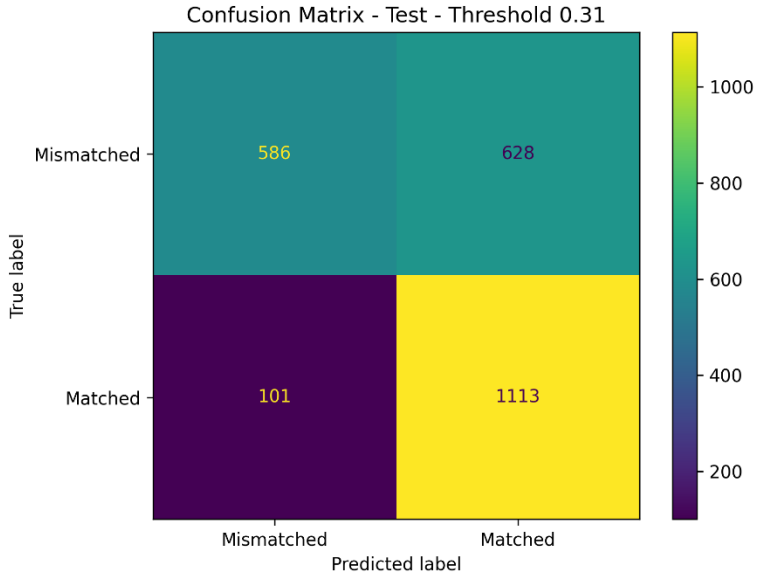
• Precision ≈ 0.6393

• Recall ≈ 0.9168

• F1-score ≈ 0.7533

• ROC-AUC ≈ 0.7766

The confusion matrix shows high recall for the Matched class, but also a relatively high number of false positives. Thus, the model identifies truly related pairs well, but sometimes incorrectly classifies unrelated pairs as matching.



	precision	recall	f1-score	support
Mismatched	0.85	0.48	0.62	1214
Matched	0.64	0.92	0.75	1214
accuracy			0.70	2428
macro avg	0.75	0.70	0.68	2428
weighted avg	0.75	0.70	0.68	2428

6.6 Error Analysis

Error analysis showed that false positives often occur when the image and caption share superficial vocabulary or objects but differ in their precise meaning. False negatives also revealed that some genuinely related pairs, especially those involving more complex images or more specific captions, are rejected by the model. This reflects the limitations of using independent embeddings and a linear classifier for a problem that is fundamentally and deeply multimodal.

One example of a false negative:

True label: 1

Predicted label: 0

Match probability: 0.02307621651202904

Actual image captions:

- A man in sunglasses , blue shirt and beige pants walks across an athletic field .
- A man is walking along a field .
- A man walks down a field with a crowd in the stands behind him .

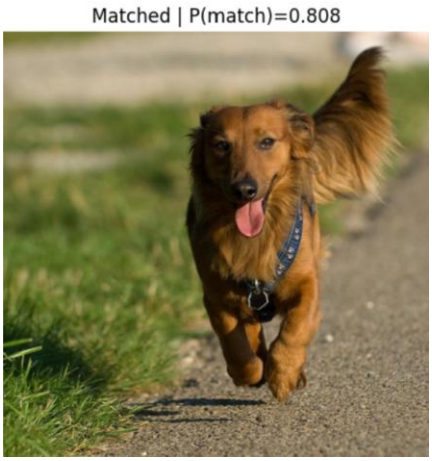
Candidate text captions:

- A man in sunglasses , blue shirt and beige pants walks across an athletic field .
- A man is walking along a field .
- A man walks down a field with a crowd in the stands behind him .



6.7 Practical Example

An arbitrary input example was also provided to the model; it classified the image-caption pair as a match with probability 0.807992. This shows that the second model can serve as a basic tool for identifying image-text matches.



7. Third Model: Image Caption Generation

7.1 Objective of the Third Model

The third approach went beyond analyzing alignment alone and turned the task into natural language generation from an image. This is the project's main generative component, testing whether a compact visual representation can predict a sequence of words.

7.2 Model Architecture

To avoid the computational burden of training a CNN from scratch, pretrained ResNet50 was used as an encoder; its 2,048-dimensional output for each image was cached. The decoder was designed as a GRU. This choice offered simplicity, fast training, and compatibility with the project's time and hardware constraints.

7.3 Vocabulary and Sequences

The vocabulary was constructed exclusively from the training set to prevent leakage. Previous analysis indicated that min_freq=5 provides a good balance between removing rare words and retaining textual coverage. The final vocabulary therefore contained 2,482 words. MAX_LEN=20 was chosen for sequences, covering approximately 95% of training captions.

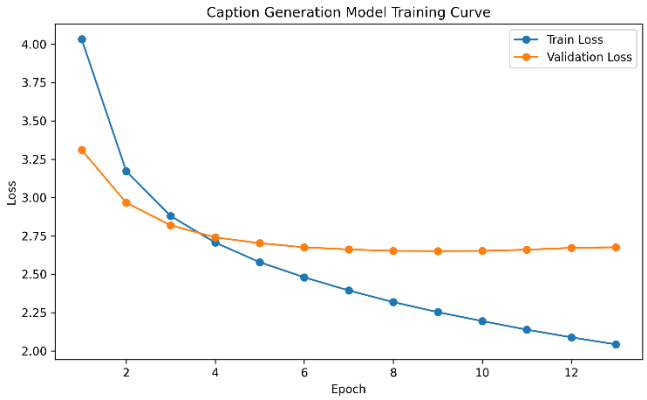
Sample caption: a black dog and a spotted dog are fighting

Encoded ids: [1, 4, 214, 607, 43, 4, 2018, 607, 58, 746, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0]

Decoded tokens: ['<start>', 'a', 'black', 'dog', 'and', 'a', 'spotted', 'dog', 'are', 'fighting', '<end>', '<pad>', '<pad>', '<pad>', '<pad>', '<pad>', '<pad>', '<pad>', '<pad>', '<pad>']

7.4 Training and Overfitting

The model was trained with cross-entropy loss and early stopping. The best epoch was 9, with a best validation loss of 2.6513. After epoch 9, training loss continued to decrease, but validation loss no longer improved. Slight overfitting was therefore observed, and early stopping with patience=4 was activated. This behavior clearly demonstrates the need to control model capacity with a relatively limited dataset.



7.5 Quantitative Results

Evaluation results for the caption-generation model on the test set were:

• BLEU-1 = 0.5764

• BLEU-2 = 0.3889

• BLEU-3 = 0.2456

• BLEU-4 = 0.1634

• ROUGE-L = 0.4514

• Best validation perplexity = 14.17

BLEU-1 and ROUGE-L indicate reasonable performance in identifying key words and their relative order. However, decreasing BLEU scores for higher-order n-grams reveal weaknesses in generating more exact, longer phrases.

	Metric	Value
0	Best validation loss	2.651252
1	Best validation perplexity	14.171767
2	BLEU-1	0.576444
3	BLEU-2	0.388900
4	BLEU-3	0.245554
5	BLEU-4	0.163389
6	ROUGE-L	0.451441

7.6 Qualitative Analysis

Successful examples showed that the model can generate meaningful captions close to the references. For instance, for an image of a dog in grass, it produced "a dog runs through the grass," which exactly matched one of the references. In other cases, however, the model generated generic or incorrect captions, such as attributing a sporting action or a different scene to an image that did not contain it. These errors indicate a bias toward common patterns, as well as the limitations of a GRU without attention.

An example of a correct prediction:



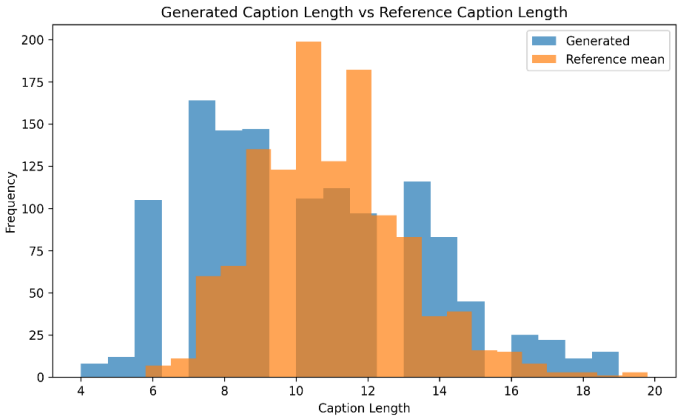
```
================================================================================
Image: 3254817653_632e840423.jpg
Generated: a group of people are standing in front of a brick building
References:
1. A family picture
2. A group of adults wear sweaters and have their hands in their pockets .
3. A group of five adults stand posing in front of buildings .
4. A group of people stand together and pose in front of a house .
5. A group shot of four guys and one girl in front of a brick building .
```

An example of an incorrect prediction:

```
Generated: three men playing basketball
BLEU-1=0.000 | BLEU-4=0.000 | ROUGE-L=0.000
References:
  1. a group of women are performing onstage
  2. people in a dance
  3. some women in white perform on a stage
  4. women participate in a skit onstage
  5. women who look alike are performing on a stage
```

Worst Caption Generation Examples #1





	generated_length	mean_reference_length
count	1214.000000	1214.000000
mean	10.229819	10.879736
std	3.184992	2.090823
min	4.000000	5.800000
25%	8.000000	9.400000
50%	10.000000	10.800000
75%	13.000000	12.000000
max	19.000000	19.800000

7.7 Summary of the Third Model

The caption-generation model showed that a compact visual representation can be used to produce a textual description, but generation quality still depends on the decoder architecture and the strength of image-text alignment. Accordingly, the simple ResNet50+GRU model is a suitable starting point, while attention-based or Transformer-based architectures are recommended for higher quality.

8. Comparison of the Three Models

The three designed approaches differed substantially in their objectives, inputs, outputs, type of learning, and evaluation metrics. The comparison is summarized below:

• Model 1: Analytical, unsupervised/semi-supervised; designed to understand semantic structure and image-text alignment.

• Model 2: Supervised binary classification; designed to determine match/mismatch for image-caption pairs.

• Model 3: Supervised and generative; designed to generate a caption from an image.

In terms of outputs, Model 1 provided the greatest analytical value, Model 2 offered the clearest classification evaluation, and Model 3 came closest to the original captioning problem.

In terms of performance:

• Model 1 showed that alignment between image and text spaces is limited but meaningful.

• Model 2 achieved acceptable matching performance with F1≈0.753 and ROC-AUC≈0.777.

• Model 3 generated meaningful captions with BLEU-1≈0.576 and ROUGE-L≈0.451.

In terms of complexity and generalization, Models 1 and 2 were lighter, while Model 3 depended most heavily on decoder capacity and vocabulary quality. Nevertheless, Model 3 is the most practically important part of the project, because it directly uses one modality to predict the other.

9. Bias and Variance Analysis

Bias and variance analysis is particularly important for the caption-generation model. Training and validation loss curves showed that both losses decreased through epoch 9, but training loss continued to fall afterward while validation loss no longer improved. This pattern indicates mild overfitting.

In summary:

• If both training and validation loss remain high, the model is underfitting.

• If training loss is low but validation loss is high, the model is overfitting.

• If both are low and close together, the model generalizes better.

In this project, the third model was reasonably balanced, but signs of overfitting emerged after epoch 9. Early stopping prevented the problem from worsening. In the matching model, the considerable number of false positives also shows that simple Logistic Regression is influenced by superficial similarities, representing another limitation in generalization.

10. Limitations and Future Suggestions

10.1 Limitations

• Separate visual and textual features: In Models 1 and 2, image and text representations were extracted independently rather than learned jointly.

• Limitations of TF-IDF: This method does not fully understand deep semantic relationships or paraphrases.

• Simplicity of the matching model: Logistic Regression is overly simple for a complex multimodal task.

• Absence of attention in caption generation: A GRU without attention may fail to associate all image components with the correct words.

• Limitations of automatic metrics: BLEU and ROUGE do not capture every semantic aspect.

10.2 Future Suggestions

• Use contrastive multimodal models such as CLIP.

• Add attention or a Transformer decoder for caption generation.

• Apply hard negative mining to the matching model.

• Use human evaluation or more advanced semantic metrics.

• Use beam search to improve the caption-generation model output.

11. Conclusion

This project on Flickr8k showed that multimodal data can be analyzed from three perspectives: structural similarity analysis, image-text matching, and caption generation. The EDA results revealed a long-tailed vocabulary distribution and showed that, with five captions per image, the dataset provides a suitable basis for multimodal modeling.

The first model established that the image and text spaces are not fully aligned, despite meaningful semantic overlap. The second showed that even a linear classifier with combined features can identify image-text matches with acceptable performance. The third showed that ResNet50 visual features can generate meaningful captions, although the final quality was limited by the simplicity of the decoder.

Overall, the project demonstrates that multimodal data analysis requires a combination of EDA, feature design, multi-metric evaluation, and critical analysis. Beyond a simple experiment, it provides a scientific framework for understanding the relationship between images and text.

12. References

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13. Code Appendix

The complete implementation code is supplied in the appendix as a Notebook, together with its outputs, plots, and intermediate files. All cells contain comments.